

Rohan Khatavkar

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EDUCATION

Arizona State University
PhD in Mechanical Engineering
CGPA: 4/4

Dec 2028 (Expected)

College of Engineering Pune, Savitribai Phule Pune University
Master of Technology in Mechanical Engineering
CGPA: 9.51/10

Aug 2021

ACADEMIC RESEARCH EXPERIENCE

Graduate Research Associate, SEMTE,
Arizona State University
Advisor: [Dr. Jiefeng Sun](#), and [Dr. Hyunglae Lee](#)

Jan 2024 - present

Soft Tunable Passive Back Support Device

- Developed a soft passive back support device that can provide task-oriented assistance by tuning the stiffness (0.58 N/mm, 0.92 N/mm, and 1.7 N/mm) and slack length (0 to 67 mm).
- The hybrid device design provides assistance tuning in a passive device without making the device heavy and bulky like the state-of-the-art tunable active devices.
- Showed the device's efficacy in assisting humans regain balance after slip perturbations while standing and walking through a biomechanics study.

Soft Semi-Active Back Support Device

- Developed a soft semi-active back support device by combining a variable stiffness passive element with a pneumatic artificial muscle in parallel enabling external energy addition to the human body with a compact and lightweight device.
- Showed human back muscle effort reduction with device use via an electromyography study.

Research Fellow, Centre for Biomedical Engineering,
Indian Institute of Technology Delhi
Advisor: [Dr. Deepak Joshi](#)

October 2021 - October 2023

Haptic Cueing System for Parkinson's Disease

- Formulated a novel kinematic gait parameter driven vibrotactile cueing system for mitigation of freezing of gait and enhancement of toe clearance in persons with Parkinson's disease (PD).
- Pilot tested the cueing system on ten people with PD. Prepared manuscripts presenting the significantly reduced freezing severity with the cueing system in 8 out of 10 persons with PD and increased toe clearance in 7 out of 8 persons with PD.

Force Myography-based User Intent Detection

- Developed force-myography (FMG) bands for step length change intent detection in amputees.

- The decision tree-based algorithms using FMG signal detected step length change with an F1 score of 86.64 percent.

Cable-driven Knee Exoskeleton

- Built a cable driven knee exoskeleton for classification of persons with osteo arthritis based on Kellgren and Lawrence grades.

Powered Ankle Foot Prosthesis

- Developed a powered ankle foot prosthesis for assisting amputees in weight shifting exercises during the early post amputation rehabilitation.

Master's Thesis, College of Engineering Pune

Advisor: [Dr. Ajay Bhattu](#)

August 2020 - August 2021

Synthesis and Fatigue Life Optimization of a Spring-less Compliant Robot Leg Design

- Exploited inherent elastodynamics of compliant mechanisms to develop a spring-less compliant quadruped robot leg design. Used Abaqus CAE for the numerical evaluation of the kinematics and dynamics of the complaint design.
- The compliant design simplified the leg assembly significantly by reducing the number of parts from 5 to 2 while maintaining a kinematic and dynamic equivalence with the spring-loaded rigid linkage.

INDUSTRIAL RESEARCH AND DEVELOPMENT EXPERIENCE

R&D Intern, Garrett Motion

August 2021 - September 2021

Torque to yield Bolted Joints for High Pressure Compressors

- Designed a joint using torque-to-yield bolt to maintain joint integrity at high temperatures.

R&D Intern, American Axle and Manufacturing

August 2017 - May 2018

Electronic Differential Locking Mechanism

- Incorporated an electronic locking feature to an existing open differential making the least possible changes to the existing assembly and to build a functional prototype.
- Developed an innovative concept of assembling the electromagnetic actuator and the locking mechanism on opposite sides of the ring gear. This enabled the modified differential case assembly packaging within the existing compact carrier and axle housing.
- Used Hypermesh and Abaqus solver for the FEA of the modified and new components. Used selective laser sintering (additive manufacturing) for prototyping.
- Formulated an algorithm for automatic locking and unlocking of the differential without driver intervention using MATLAB and Simulink.

R&D Staff, College Racing Team

August 2015 - October 2016

SAE EffiCycle 2016, a competition for which college teams design and fabricate a three-wheeler that is both motor and manually driven:

- Designed and developed a chain and cassette transmission and a reduction gearbox, and an innovative differential gearbox using ratchets.
- Calculated drive train performance characteristics. Modeled drive train elements, worked on vehicle assembly, and was responsible for drafting (including GD&T) of knuckles for water jet cutting on SolidWorks.

SAE Supra 2016, a competition for which college teams design and fabricate a race car:

- Designed and developed the engine intake manifold.
- Simulated airflow through a venturi and orifice to design airflow restriction in the intake manifold using Abaqus CAE.

AWARDS AND FELLOWSHIPS

Jumpstart Research Grant at Arizona State University	\$ 1500	Spring and Fall 2025
Senior Research Fellowship at Indian Institute of Technology Delhi	\$ 10,895.92	Oct 2021 - Sep 2023
All India Council for Technical Education (AICTE) Post Graduate Scholarship	\$ 3386.2	Aug 2019 - Jul 2021
Bravo Award at Garrett Motion		September 2021
Secured 98.22 percentile in a highly competitive national master's entrance exam (GATE).		March 2019

PUBLICATIONS

[1] **Khatavkar, Rohan**, The Bach Nguyen, Yuanhao Chen, Hyunglae Lee, and Jiefeng Sun. "A Hybrid Variable-Stiffness Soft Back Support Device." In *IEEE Robotics and Automation Letters*, (2025). <https://doi.org/10.1109/LRA.2025.3541459>

[2] Weissman, Eric, **Rohan Khatavkar**, Jiefeng Sun. "A Well-Tuned HARP: Versatile Artificial Muscle by Decoupling Anisotropy. " In *Proceedings of the National Academy of Sciences of the United States of America (PNAS)*, 2025. (under review)

[3] **Khatavkar, Rohan**, The Bach Nguyen, Inseung Kang, Hyunglae Lee, and Jiefeng Sun. "A Semi-Active Soft Back Support Device." In preparation, (2026).

[4] **Khatavkar, Rohan**, Ashutosh Tiwari, Priyanka Bhat, Achal Srivastava, S. Senthil Kumaran, and Deepak Joshi. "A Novel Kinematic Gait Parameter-Based Vibrotactile Cue for Freezing of Gait Mitigation Among Parkinson's Patients: A Pilot Study." In *IEEE Transactions on Haptics*, 2024. <https://doi.org/10.1109/TOH.2024.3378917>

[5] **Khatavkar, Rohan***, Ashutosh Tiwari*, Priyanka Bhat, Deepak Joshi. "Effect of a Kinematic Gait Parameter-Based Haptic Cue on Toe Clearance in Parkinson's Patients." In *Annals of Biomedical Engineering*, 2024. <https://doi.org/10.1007/s10439-024-03501-4>

[6] **Khatavkar, Rohan Vijay**, and Ajay Pandit Bhattu. "Synthesis and Fatigue Life Optimization of a Spring-Less Compliant Robot Leg Design." In *National Conference on Multidisciplinary Analysis and Optimization*, pp. 451-459. Springer, Singapore, 2023. https://doi.org/10.1007/978-981-19-3938-9_48

[7] **Khatavkar, Rohan**, Ashutosh Tiwari, Rishabh Bajpai, and Deepak Joshi. "Gait Step Length Classification Using Force Myography." In *2022 International Conference for Advancement in Technology (ICONAT)*, pp. 1-4. IEEE, 2022. <https://doi.org/10.1109/ICONAT53423.2022.9726014>

[8] Tiwari, Ashutosh, Rishabh Bajpai, **Rohan Khatavkar**, Manish Gupta, Bhavuk Garg, and Deepak Joshi. "Exploring the Center of Pressure Shift Feedback at Heel Strike to Modulate the Step Length."

In 2022 *International Conference for Advancement in Technology (ICONAT)*, pp. 1-5. IEEE, 2022.
<https://doi.org/10.1109/ICONAT53423.2022.9726031>

[9] Bajpai, Rishabh, Ashutosh Tiwari, Deepak Joshi, and **Rohan Khatavkar**. "AbnormNet: A Neural Network Based Suggestive Tool for Identifying Gait Abnormalities in Cerebral Palsy Children." In 2022 *International Conference for Advancement in Technology (ICONAT)*, pp. 1-5. IEEE, 2022.
<https://doi.org/10.1109/ICONAT53423.2022.9725832>

SKILLS

Tools: MATLAB, PyTorch, OpenSim, Arduino, Raspberry Pi, Vicon Nexus, SolidWorks, Fusion 360, Siemens NX, Abaqus CAE, Ansys, SMC PneuDraw, Overleaf, Adobe Illustrator, Adobe Premiere Pro.

Design: Mechatronics system design, robotic system design, pneumatic system design, CAD, additive manufacturing, solid mechanics, FEA.

Experimental Skills: Force testing (Instron), optical motion capture (Vicon), EMG (Delsys) acquisition and processing, multi-sensor data capture in real-time target machines (Speedgoat), movement biomechanics analyses, signal processing, clinical experimental protocol formulation, patient testing.

EXTRACURRICULAR

Graduate Assistant at College of Engineering Pune: Assisted the Admissions Committee in evaluating applicants to the master's program in mechanical engineering.

PROFESSIONAL SERVICE

Reviewer for:

Scientific Reports
IEEE Sensors Journal
IEEE Robotics and Automation Letters
IEEE RoboSoft
IEEE/ASME AIM
American Control Conference

PROFESSIONAL MEMBERSHIP

Member, Tau Beta Pi
Student Member, IEEE